

Muhammad Talha

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Portfolio

Summary

Computer Science graduate with practical experience in machine learning, backend development, and AI applications. Skilled in building end-to-end ML pipelines, deploying models with FastAPI and Docker. Interested in an entry-level Machine Learning, Data Science, or Software Engineering role.

Experience

- Data Science Trainee, Machine Learning 1** June 2026 - July 2026
Built a binary classification model to predict heart disease risk from patient health indicators. Performed exploratory data analysis and compared multiple algorithms, including Logistic Regression, SVM, KNN, and Decision Tree. Achieved 87% accuracy using Logistic Regression through hyperparameter tuning and cross-validation.
Tools: Python, Scikit-learn, Pandas, NumPy, Matplotlib

Projects

- Next-Gen AI Desktop Assistant (Final Year Project)** Gen AI / Agentic Systems
Developed an offline AI desktop assistant capable of voice and text interaction using local LLMs. Implemented modular task execution, file management, application control, and long-term memory using ChromaDB with privacy-preserving on-device inference.
Tools: Python, Ollama, Gemma 3, ChromaDB, PySide6, WhisperFlow, NLTK
- Insurance Premium Category Predictor** ML Engineering / Backend
Built and deployed a production-ready ML classification service via FastAPI and Docker, achieving consistent cross-environment performance. Designed structured logging and debugging pipelines that reduced error diagnosis time significantly. Containerized the full inference stack for reproducible deployments on Linux servers.
Tools: FastAPI, Docker, Scikit-learn, Python, Linux
- Multilingual Sentiment Analysis (Urdu & English)** NLP / Transformers
Fine-tuned a multilingual transformer model (mBERT) on a custom Urdu-English sentiment dataset using Hugging Face. Implemented a custom tokenization pipeline to handle code-switched text and achieved measurable F1-score improvements over baseline models.
Tools: Hugging Face Transformers, PyTorch, mBERT

Research Experience

- Research Paper: Beyond the Cloud (Undergraduate Research)** 2026
Co-authored a research paper as part of the undergraduate Final Year Project. Proposed an offline-first AI desktop assistant featuring local LLM inference, speech recognition, modular command routing, and privacy-by-design architecture. Conducted literature review, system design, implementation, experimental evaluation, and technical documentation.
Keywords: Local LLMs, Edge AI, NLP, Speech Recognition, Privacy-Preserving AI, Desktop Automation

Education

- University of Engineering and Technology, Lahore** 2022 – 2026
Bachelor of Science in Computer Science

Technical Skills

- Programming:** Python, SQL, C/C++, HTML, CSS
- Machine Learning :** Scikit-learn, PyTorch, Transformers, NLP
- Backend:** FastAPI, REST APIs, Pydantic
- Data:** Pandas, NumPy, Feature Engineering, EDA
- Tools:** Docker, Git, Linux, VS Code, Jupyter Notebook, Google Colab

Certifications

- Machine Learning Specialization – Stanford / DeepLearning.AI (Coursera)** 2025
- Intermediate SQL – DataCamp** 2025